BRADY ADAMS

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EDUCATION

Indiana University Bloomington | Bloomington, IN

B.S. Computer Engineering • Expected May 2027

Minor in Intelligent Systems Engineering

RESEARCH & ENGINEERING PROJECTS

EnerGaia — Powering Science Education Through Gamified Energy Systems | TechTeach Innovation Research Project, 2026

- Collaborated on an IU-funded interdisciplinary research project with 3 Intelligent Systems Engineering students, 2 Education students, and 1 Luddy graduate student to investigate how engineering and education disciplines can jointly develop technology-enhanced science learning experiences.
- Developed a Unity 2D educational game in C# for middle school science that teaches Potential and Kinetic Energy through physics-based movement, interactive energy mechanics, checkpoint instruction, and quiz-driven boss encounters.
- Implemented core gameplay systems including player movement, PE/KE mechanics, equipment-based abilities, checkpoints, dialogue, scene transitions, boss health and stun phases, dynamic quizzes, and teacher assessment tools.
- Applied Universal Design for Learning (UDL) principles through visual and textual instructional supports, accessible color/texture choices, multiple quiz interaction methods, and immediate gameplay feedback.
- Developed a teacher dashboard for custom question entry and student performance review, allowing educators to use the prototype as an assessment tool.
- Presented the research and prototype at the HASTI Mini-Conference, an IU research conference in Alumni Hall, and directly to an 8th-grade science classroom with a live game demonstration.

Embedded Machine Learning for Motion Detection | Raspberry Pi / TensorFlow Lite

- Collaborated on development and deployment of a lightweight CNN for real-time motion classification using gyroscope and accelerometer data on a Raspberry Pi; optimized through windowing, pruning, and quantization to reduce model size by over 66% and improve accuracy from 92% to 95%.
- Co-authored and presented a research poster at a public academic symposium, explaining the technical approach, optimization methods, and results.

Raspberry Pi Board Game | Python / Embedded Systems

- Built an interactive board game using a Raspberry Pi, Python, an 8×8 display, 3D-printed game pieces, and a laser-cut board; programmed answer randomization and visual output.

TECHNICAL SKILLS

Languages: Python, C#, C/C++, Verilog/SystemVerilog **ML/AI:** TensorFlow, TensorFlow Lite, scikit-learn, CNNs, Random Forest, PCA, quantization, pruning **Development:** Unity, REST APIs, Linux, Docker **Embedded:** Raspberry Pi, sensor data processing, Vivado

EXPERIENCE

Flow Space Movement Center | Avon, IN

Certified Parkour / Gymnastics Instructor • Oct. 2019 – Aug. 2025

- Taught parkour to students of varied ages and abilities; developed lesson plans, assessed performance, provided individualized feedback, and coached safe technique.

Tropical Smoothie Cafe | Avon, IN

Retail Associate • Jun. 2023 – Aug. 2023

- Provided customer service, assisted with product selection and recommendations, answered questions, and maintained store appearance.